

Introduction to SG Math

Base-Ten Blocks, Bar Models & Mental Math Strategies

Part 1: Friendly Numbers

Example 1 (Compatible Numbers): Solve $128 + 245 + 273 + 155 + 27 + 72$ by grouping friendly pairs.

① **Identify compatible pairs**

Look at the ones and tens digits to pair numbers that make whole hundreds:

$$\boxed{128} + \boxed{245} + \boxed{273} + \boxed{155} + \boxed{27} + \boxed{72}$$

② **Rearrange and group the pairs**

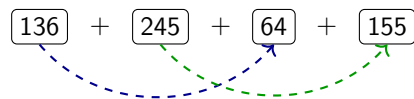
$$(128 + 72) + (245 + 155) + (273 + 27)$$

③ **Add the friendly round hundreds**

$$200 + 400 + 300 = \mathbf{900}$$

Practice Questions

1. **Pair compatible numbers to make hundreds, then solve:**

$$\boxed{136} + \boxed{245} + \boxed{64} + \boxed{155}$$


Equation: $(\underline{\hspace{1cm}} + \underline{\hspace{1cm}}) + (\underline{\hspace{1cm}} + \underline{\hspace{1cm}}) = \underline{\hspace{1cm}} + \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$

2. **Rearrange into three compatible pairs to solve $118 + 265 + 234 + 135 + 66 + 82$:**

Equation: $(\underline{\hspace{1cm}} + \underline{\hspace{1cm}}) + (\underline{\hspace{1cm}} + \underline{\hspace{1cm}}) + (\underline{\hspace{1cm}} + \underline{\hspace{1cm}}) = \underline{\hspace{1cm}} + \underline{\hspace{1cm}} + \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$

3. **Rearrange into compatible pairs to solve $345 + 182 + 155 + 218$:**

Equation: $(\underline{\hspace{1cm}} + \underline{\hspace{1cm}}) + (\underline{\hspace{1cm}} + \underline{\hspace{1cm}}) = \underline{\hspace{1cm}} + \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$

Part 2: Rounding and Compensation

Example 2 (Rounding & Compensation): Solve $999 + 99 + 9$ using friendly benchmark numbers.

① **Round up to the nearest powers of 10**

Each number is just 1 less than a round benchmark (1000, 100, 10):

$$\boxed{999} \xrightarrow{+1} \boxed{1000} \quad \boxed{99} \xrightarrow{+1} \boxed{100} \quad \boxed{9} \xrightarrow{+1} \boxed{10}$$

② **Add the round numbers and track the total excess**

We added $1 + 1 + 1 = 3$ extra, so we subtract 3 from the round total:

$$(1000 + 100 + 10) - 3 = 1110 - 3$$

③ **Calculate the final sum**

$$1110 - 3 = 1107$$

Practice Questions

1. Use rounding and compensation to calculate $499 + 199 + 19$:

$$\boxed{499} \xrightarrow{+1} 500 \quad \boxed{199} \xrightarrow{+1} 200 \quad \boxed{19} \xrightarrow{+1} 20$$

Equation: $(\underline{\hspace{1cm}} + \underline{\hspace{1cm}} + \underline{\hspace{1cm}}) - \underline{\hspace{1cm}} = \underline{\hspace{1cm}} - \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$

2. Use rounding and compensation to calculate $599 + 299 + 89$:

Equation: $(\underline{\hspace{1cm}} + \underline{\hspace{1cm}} + \underline{\hspace{1cm}}) - \underline{\hspace{1cm}} = \underline{\hspace{1cm}} - \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$

3. Use rounding and compensation to calculate $398 + 198 + 98$ (round each up by +2):

Equation: $(\underline{\hspace{1cm}} + \underline{\hspace{1cm}} + \underline{\hspace{1cm}}) - \underline{\hspace{1cm}} = \underline{\hspace{1cm}} - \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$

Part 3: Shifting (Subtraction ONLY)

CRITICAL RULE: The Shifting Method ONLY works on subtraction!

In subtraction, adding or subtracting the **same amount** to both numbers preserves the constant difference ($A - B = (A + c) - (B + c)$). **Do NOT use this for addition!**

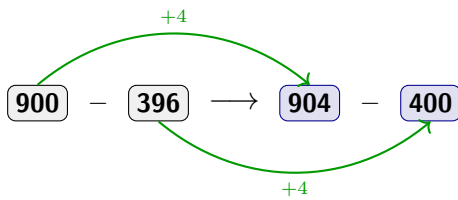
Example 3 (The Shifting Method for Subtraction): Solve $900 - 396$ by shifting to a round number.

① **Identify the nearest round hundred**

396 is only 4 away from 400.

② **Add the same amount (+4) to both numbers**

Adding the same number to both terms preserves the difference (no borrowing across zeroes!):

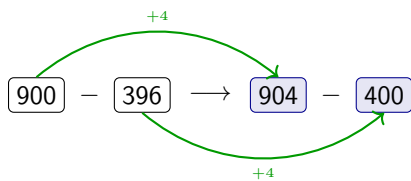


③ **Subtract without regrouping**

$$900 - 396 = 904 - 400 = \mathbf{504}$$

Practice Questions

Use the shifting method (constant difference) to make a round subtrahend:



1. $900 - 396 = (\underline{\hspace{2cm}} + 4) - (396 + 4) = 904 - 400 = \underline{\hspace{2cm}}$

2. $800 - 297 = (800 + \underline{\hspace{2cm}}) - (297 + \underline{\hspace{2cm}}) = \underline{\hspace{2cm}} - 300 = \underline{\hspace{2cm}}$

3. $600 - 189 = (600 + \underline{\hspace{2cm}}) - (189 + \underline{\hspace{2cm}}) = \underline{\hspace{2cm}} - \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

4. $1000 - 495 = (1000 + \underline{\hspace{2cm}}) - (495 + \underline{\hspace{2cm}}) = \underline{\hspace{2cm}} - 500 = \underline{\hspace{2cm}}$